

Igor Lima Rocha Azevedo

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EDUCATION

• Imperial College London

2025-26

Master Research (MRes) – Artificial Intelligence and Machine Learning

- Researching GPT-style autoregressive transformer models for time series forecasting and next-day event prediction.

• University of Brasilia

2016-22

Electrical Engineering

- Focused on embedded systems, FPGAs, and Machine Learning for codec optimization and financial markets.

RESEARCH EXPERIENCE

• Research Scholar at The University of Tokyo

Tokyo, Japan | Apr 2023 – Apr 2025

- Collaborated with Nikkei Inc. (a Financial Times group company) to design novel recommender system models.
- Researched news recommender systems to address popularity bias and news avoidance supervised by Professor Toyotaro Suzumura (The University of Tokyo) and Dr. Yuichiro Yasui (Nikkei Inc.) .
- Developed deep learning models for high-frequency stock price forecasting.
- Researched foundational LLMs, exploring retrieval-augmented generation (RAG) and long-context handling.

• Research Intern at Cellcrypt: Quantum-Safe Encrypted Calls

London, UK (Remote) | Sep 2020 – Jun 2021

- Optimized machine learning models to improve VoIP performance by enhancing call quality and reducing latency.
- Built an ML pipeline for automated PJSIP parameter tuning, which boosted call quality metrics by 7%.

PROFESSIONAL EXPERIENCE

• Developer & Development Team Lead

July 2021 – April 2023

VOGA – The largest financial ecosystem in the Central-West region of Brazil.

Brasilia, Brazil

- Led system integration for VOGA post-acquisition by BTG Pactual, South America's largest investment bank.
- Architected and developed a centralized platform to monitor and track over USD 300 million in managed assets.
- Architected, deployed, and maintained a full-stack Flask/NextJS application on AWS, utilizing Terraform for infrastructure as code (ECS, RDS, ELB), Jenkins for CI/CD, and Cloudflare for security and CDN.

SELECTED PUBLICATIONS

• A Look Into News Avoidance Through AWRS: An Avoidance-Aware Recommender System

SIAM International Conference on Data Mining (SDM'25)

Paper

- Collaboration: Toyotaro Suzumura (The University of Tokyo) and Dr. Yuichiro Yasui (Nikkei Inc.)
- Highlights: Developed AWRS, a recommender system that uses article avoidance to improve news recommendations. It outperformed existing methods on English, Norwegian, and Japanese (Nikkei Inc.) datasets.

• NewsReX: A More Efficient Approach to News Recommendation with Keras 3 and JAX

Under Review

Paper

- Collaboration: Toyotaro Suzumura (The University of Tokyo) and Dr. Yuichiro Yasui (Nikkei Inc.)
- Highlights: A modular framework for news recommendation research implementing SOTA models. Optimized with Keras 3 + JAX for enhanced performance via JIT compilation and XLA.

AWARDS

• Society for Industrial and Applied Mathematics (SIAM) Travel Award – SDM25

May 2025 | Certificate

- Recognizing promising early-career researchers in the field of data mining.

• Japanese Government (MEXT) Research Scholarship

April 2023 – April 2025 | Certificate

- A prestigious scholarship to support students conducting research at higher education institutions in Japan.

• Brazilian Government (CNPq) Scientific Initiation Scholarship (PIBIC)

August 2019 – July 2020 | Certificate

- A federal government-funded program to support undergraduate students engaging in research and innovation.

LANGUAGES & SKILLS

• Languages: Native Portuguese, Fluent English, Limited Spanish, Basic Japanese

• Technical Skills: Python, PyTorch, Lightning AI, Keras, JAX, CrewAI, DSPy, Flask, FastAPI, Java, C, Nginx, SQL, AWS (EC2, ECS, RDS, Dynamo, IAM, Lambda, VPC), Terraform, Cloudflare.